

DEFINITION — HOW DATA IS DESCRIBED

CONTRACTS6

ADOPT


ODCS
BITOL @ LF

ADOPT


OpenAPI
LF

ADOPT


AsyncAPI
LF

ADOPT


GraphQL
LF

SITUATIONAL


gRPC
CNCF

SITUATIONAL


OData
OASIS

DATA PRODUCTS4

ADOPT


ODPS
BITOL @ LF

ASSESS


DPDS
ODM

ASSESS


ODPSpec
LF

ASSESS


DPROD
OMG

SCHEMA7

ADOPT


XML Schema
W3C

ADOPT


JSON Schema
IETF

ADOPT


SQL DDL
ISO/IEC

ADOPT


AVRO Schema
ASF

ADOPT


Protobuf
GOOGLE

ASSESS


LinkML
LINKML

ASSESS


Table Schema
FRictionless

SEMANTICS8

SITUATIONAL


RDF/OWL
W3C

SITUATIONAL


DCAT
W3C

SITUATIONAL


SKOS
W3C

SITUATIONAL


SHACL
W3C

SITUATIONAL


JSON-LD
W3C

ASSESS


OSI
OSI INITIATIVE

ASSESS


schema.org
W3C

ASSESS


ShEx
W3C

STORAGE — WHERE DATA LIVES

FILE FORMATS8

ADOPT


CSV
IETF

ADOPT


JSON
IETF

ADOPT


XML
W3C

ADOPT


YAML
YAML.ORG

ADOPT


PARQUET
ASF

ADOPT


AVRO
ASF

SITUATIONAL


ORC
ASF

ASSESS


Lance
LANCE

OPEN TABLE FORMATS4

ADOPT


Iceberg
ASF

SITUATIONAL


Delta
LF

ASSESS


Hudi
ASF

ASSESS


Lance
LANCE

STORAGE SYSTEMS2

ADOPT


S3
AWS

CAUTION


HDFS
ASF

MOVEMENT — HOW DATA FLOWS BETWEEN SYSTEMS

DATABASE CONNECTIVITY4

ADOPT


JDBC
JCP

ADOPT


ODBC
ISO/IEC

ADOPT


ADBC
ASF

CAUTION


XMLA
MICROSOFT

INTERCONNECTION3

ADOPT


HTTP
IETF

SITUATIONAL


Delta Sharing
LF

CAUTION


FTP / SFTP
IETF

MESSAGING5

ADOPT


Kafka
ASF

ADOPT


CloudEvents
CNCF

SITUATIONAL


MQTT
OASIS

SITUATIONAL


AMQP
OASIS

CAUTION


JMS
JAKARTA EE

TRANSFORMATION — HOW DATA IS PROCESSED AND RESHAPED

IN-MEMORY FORMAT2

ADOPT


DataFrame API
DATA-APIs

ADOPT


Apache Arrow
ASF

PROCESSING7

ADOPT


Spark
ASF

ADOPT


Pandas
NUMFOCUS

ADOPT


SQL DML
ISO/IEC

ADOPT


dbt
DBT LABS

SITUATIONAL


Beam
ASF

ASSESS


Ibis
IBIS

CAUTION


XSLT
W3C

DISCOVERY — HOW DATA IS FOUND AND TRACED

CATALOG APIS5

ADOPT


Iceberg Catalog
ASF

ADOPT


Schema Registry
CONFLUENT

SITUATIONAL


Unity Catalog
LF

ASSESS


DuckLake
DUCKDB LABS

CAUTION


Hive Metastore
ASF

LINEAGE2

ADOPT


OpenLineage
LF

ASSESS


PROV
W3C

OPERATIONS — HOW DATA IS QUERIED, OBSERVED, GOVERNED

QUERY5

ADOPT


SQL
ISO/IEC

SITUATIONAL


Substrait
LF

SITUATIONAL


SPARQL
W3C

SITUATIONAL


GQL
ISO/IEC

CAUTION


MDX
MICROSOFT

DATA QUALITY3

SITUATIONAL


Great Expectations
GX LABS

SITUATIONAL


dbt tests
DBT LABS

SITUATIONAL


SodaCL
SODA

OBSERVABILITY2

ADOPT


Open Telemetry
CNCF

ASSESS


OORS
BITOL @ LF

POLICIES2

ADOPT


OPA
CNCF

ASSESS


ODRL
W3C

AI INTERFACES2

ADOPT


MCP
LF

SITUATIONAL


A2A
LF